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Brazing probe

Abstract

A brazing apparatus to join a first component and a second component includes a heating element configured to melt a ring located at a joint between the first component and the second component, and a probe configured to contact the ring. A position of the probe is biased toward the ring. The probe is installed at a probe support. The probe is movable along a probe central axis relative to the probe support. A sensor is operably connected to the probe and is configured to determine movement of the probe along the probe central axis. The movement of the probe is indicative of melting of the ring.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/358,308 filed Jul. 5, 2022, the disclosure of which is incorporated by reference in its entirety.

BACKGROUND

(1) Exemplary embodiments pertain to the art of aluminum brazing, and more specifically to an aluminum brazing apparatus and probes suitable for manufacturing a heat exchanger.

(2) Brazing is a metal-joining process with many applications, such as the joining of tubes for heat exchangers. In the joining of aluminum heat exchanger tubes, a brazing torch is used to melt an alloy ring at the joint between a first tube and a second tube. The process has a small temperature window, such that if the torch is held at the alloy ring for too long a time the tubes may be damaged due to overexposure to the heat from the torch. Typically, the time the torch is held at the braze joint is based on a visual assessment by the operator of the torch. Reliance on the visual assessment can lead to inconsistent braze joints and damaged tubes.

BRIEF DESCRIPTION

(3) In one embodiment, a brazing apparatus to join a first component and a second component includes a heating element configured to melt a ring located at a joint between the first component and the second component, and a probe configured to contact the ring. A position of the probe is biased toward the ring. The probe is installed at a probe support. The probe is movable along a probe central axis relative to the probe support. A sensor is operably connected to the probe and is configured to determine movement of the probe along the probe central axis. The movement of the probe is indicative of melting of the ring.

(4) Additionally or alternatively, in this or other embodiments a compression spring is operably connected to the probe to bias the position of the probe toward the ring.

(5) Additionally or alternatively, in this or other embodiments the compression spring is secured at an exterior surface of the probe.

(6) Additionally or alternatively, in this or other embodiments the heating element is one of a torch, a resistive heating element or an inductive heating element.

(7) Additionally or alternatively, in this or other embodiments the probe includes one or more cooling channels having a cooling fluid circulating therethrough.

(8) Additionally or alternatively, in this or other embodiments the cooling fluid is one of air or oil.

(9) Additionally or alternatively, in this or other embodiments the probe includes and cooling fluid inlet and a cooling fluid outlet connected to the one or more cooling channels to circulate the cooling fluid through the one or more cooling channels.

(10) Additionally or alternatively, in this or other embodiments a controller is operably connected to the sensor and to the heating element. The controller is configured to stop operation of the heating element when the sensor detects movement of the probe along the probe central axis.

(11) Additionally or alternatively, in this or other embodiments the probe is formed from a metallic material.

(12) Additionally or alternatively, in this or other embodiments the probe is formed by an additive manufacturing process.

(13) In another embodiment, a method of brazing a tube includes positioning a probe in contact with a ring positioned at a joint between a first tube portion and a second tube portion. The probe is biased toward the ring. A heating element is positioned at the joint, and the ring is melted via operation of the heating element. Movement of the probe along a probe central axis is detected via a sensor operably connected to the probe, the movement of the probe indicative of melting of the ring.
(14) Additionally or alternatively, in this or other embodiments the probe is biased toward the ring via a compression spring operably connected to the probe.
(15) Additionally or alternatively, in this or other embodiments the heating element is one of a torch, a resistive heating element or an inductive heating element.
(16) Additionally or alternatively, in this or other embodiments a cooling fluid is circulated through one or more cooling channels of the probe to cool the probe.
(17) Additionally or alternatively, in this or other embodiments the cooling fluid is one of air or oil.
(18) Additionally or alternatively, in this or other embodiments operation of the heating element is stopped when the sensor detects movement of the probe along the probe central axis.
(19) Additionally or alternatively, in this or other embodiments the probe is formed from a metallic material.
(20) Additionally or alternatively, in this or other embodiments at least one of the first tube portion and the second tube portion are formed from aluminum.
(21) Additionally or alternatively, in this or other embodiments the first tube portion and the second tube portions are tubes of a heat exchanger.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:
(2) FIG. **1** is an illustration of a brazing apparatus being used for the manufacture of a heat exchanger in accordance with exemplary embodiments;
(3) FIG. **2** is an illustration of a brazing probe in accordance with exemplary embodiments;
(4) FIG. **3** is a perspective view of a brazing probe in accordance with exemplary embodiments;
(5) FIG. **4** is a cross-sectional view of a brazing probe in accordance with exemplary embodiments; and
(6) FIG. **5** is a schematic illustration of a method of brazing in accordance with exemplary embodiments.
DETAILED DESCRIPTION
(7) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.
(8) Referring to FIG. **1**, illustrated is an embodiment of a brazing apparatus **100** being used for the manufacture of a heat exchanger **10**. The heat exchanger **10** includes a plurality of tubes **12** through which a thermal energy exchange fluid, for example, refrigerant, is flowed. In some embodiments, the tubes **12** are formed from an aluminum material, while in other embodiments other metal materials may be utilized. The tubes **12** may include heat exchange tubes **12***a*, hairpin tubes **12***b*, header tubes **12***c*, and other tubes. Connections of tubes, for example, heat exchange tubes **12***a* to hairpin tubes **12***b*, is accomplished via a brazing process using a brazing apparatus **100**. The brazing apparatus **100** includes a heating element, such as a torch **14** to emit heat to melt an alloy ring **16** at a joint **60** between the heat exchange tube **12***a* and the hairpin tube **12***b*. While in the embodiments described herein, the heating element is the torch **14**, in other embodiments other heating elements, such as resistive or inductive heating elements may be utilized to melt the alloy

ring **16**. Once the alloy ring **16** is melted, the torch **14** is withdrawn to allow the alloy ring **16** to re-solidify thus joining the heat exchange tube **12***a* to the hairpin tube **12***b*. A probe **18** is placed in contact with the alloy ring **16**, and is utilized to detect melting of the alloy ring **16**. A signal is then provided to the operator of the torch **14**, that the torch **14** may be withdrawn from the alloy ring **16**. It should be appreciated that, in certain instances the signal my enable the automatic withdrawing of the torch **14** from the alloy ring **16** (i.e., without operator intervention).

(9) Referring now to FIGS. **2** and **3**, the probe **18** of the brazing apparatus **100** will now be described in more detail. The probe **18** includes a probe tip **20**, which contacts the alloy ring **16**, as shown in FIG. **1**. The probe **18** is mounted on a probe support **22** and extends through a probe sleeve **26**. In particular, the probe sleeve **26** has a sleeve opening **28** through which the probe **18** extends. The probe **18** is movable along a probe central axis **30** relative to the probe sleeve **26**, and is biased toward the alloy ring **16** by a compression spring **32** mounted on a probe exterior **34** of the probe **18** and retained to the probe **18** by, for example a retaining ring **36**. One skilled in the art will readily appreciate that the compression spring **32** may be retained at the probe exterior **34** by other means, such as a screw or the like.

(10) As stated, the probe **18** contacts and is biased toward the alloy ring **16** by the compression spring **32**. As the torch **14** applies heat to the alloy ring **16**, the alloy ring **16** melts and softens. This allows for movement of the probe **18** along the probe central axis **30** as driven by the compression spring **32**. A linear displacement sensor **38** is operably connected to the probe **18**. The linear displacement sensor **38** detects the movement of the probe **18** along the probe central axis **30** when the alloy ring **16** begins to melt, and is connected to an alarm or signal mechanism **40**, such as a light or sound, which alerts the operator of the torch **14** that the alloy ring **16** has melted, and that the operator should withdraw the torch **14** from the alloy ring **16** (as mentioned above, this withdraw of the torch **14** may be automated, removing the need for operator intervention). In other embodiments, the linear displacement sensor **38** is connected to a controller **42**. Upon receiving the signal from the linear displacement sensor **38**, the controller **42** may automatically move the torch **14** away from the alloy ring **16**, or alternatively may extinguish the torch **14** by stopping a flow of fuel to the torch **14**. While in some embodiments one probe **18** is mounted to the probe support **22**, in other embodiments as shown in FIG. **2**, multiple probes **18** are independently mounted to the probe support **22**, with each probe **18** having their own compression spring **32** and linear displacement sensor **38**. This allows for brazing at multiple locations simultaneously.

(11) Referring now to FIG. **4**, a cross-sectional view of an embodiments of a probe **18** is illustrated. The probe **18** is formed from, for example, a metal material. In some embodiments, the metal material is a stainless-steel material capable of operating at temperatures up to about 2550 degrees Fahrenheit. To improve the temperature tolerance of the probe **18**, the probe is actively cooled by, for example, air or other cooling fluid such as oil circulated through the probe **18** via one or more cooling channels **44** formed in the probe **18**. To allow for the formation of such cooling channels **44** in the probe **18**, in some embodiments the probe is formed via an additive manufacturing or 3-D printing process. In some embodiments, the cooling channels **44** extend circumferentially around the probe **18**, and the flow of cooling fluid **46** enters the probe **18** at a cooling inlet **48** and exits the probe **18** at a cooling outlet **50** after circulating through the cooling channels **44**.

(12) Referring again to FIG. **2**, in some embodiments the probe **18** is mounted to a stand **52**, such as an articulated arm. The stand **52** allows for positioning and repositioning the probe **18** as needed to point the probe **18** toward the alloy ring **16** to contact the alloy ring **16**.

(13) Referring now to FIG. **5**, a schematic of a method of brazing a first tube **12** to a second tube **12**. In step **100**, the first tube **12** and the second tube **12** are positioned and the alloy ring **16** is installed to define the joint. At step **102**, the probe **18** is positioned in contact with the alloy ring **16**, and the torch **14** is positioned at the alloy ring **16** and step **104**. The torch **14** heats the alloy ring **16** at step **106**, and the probe **18** moves along the probe central axis **30** in step **108** when the alloy ring **16** begins to melt. The movement of the probe **18** is detected by the linear displacement sensor **38**

at step **110**, and a signal indicates that the torch **14** should be moved to stop melting of the alloy ring **16**.
(14) Use of the probe **18** more allows for a more precise determination of when the alloy ring **16** begins to melt, and eliminates (or at least reduces) human factors from the determination. This helps optimize the brazing cycle time, and improves quality and consistency of the brazing operation, thus preventing damage to the tubes and preventing leakage at the joint.
(15) The term "about" is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application.
(16) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.
(17) While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A brazing apparatus suitable for joining a first component and a second component, the brazing apparatus comprising: a heating element configured to melt a ring disposed at a joint between the first component and the second component; and a probe configured to contact the ring, a position of the probe biased toward the ring, a probe support at which the probe is installed, the probe movable along a probe central axis relative to the probe support; a sensor operably connected to the probe and configured to determine movement of the probe along the probe central axis, the movement of the probe indicative of the melting of the ring; and a compression spring operably connected to the probe to bias the position of the probe toward the ring.
2. The brazing apparatus of claim 1, wherein the compression spring is secured at an exterior surface of the probe.
3. The brazing apparatus of claim 1, wherein the heating element is one of a torch, a resistive heating element, or an inductive heating element.
4. The brazing apparatus of claim 1, wherein the probe includes one or more cooling channels having a cooling fluid circulating therethrough.
5. The brazing apparatus of claim 4, wherein the cooling fluid is one of air or oil.
6. The brazing apparatus of claim 4, wherein the probe includes and cooling fluid inlet and a cooling fluid outlet connected to the one or more cooling channels to circulate the cooling fluid through the one or more cooling channels.
7. A brazing apparatus suitable for joining a first component and a second component, the brazing apparatus comprising: a heating element configured to melt a ring disposed at a joint between the first component and the second component; and a probe configured to contact the ring, a position of the probe biased toward the ring, a probe support at which the probe is installed, the probe movable along a probe central axis relative to the probe support; and a sensor operably connected to the probe and configured to determine movement of the probe along the probe central axis, the

movement of the probe indicative of the melting of the ring; and a controller operably connected to the sensor and the heating element, the controller configured to stop operation of the heating element when the sensor detects movement of the probe along the probe central axis.

8. The brazing apparatus of claim 1, wherein the probe is formed from a metallic material.

9. The brazing apparatus of claim 1, wherein the probe is formed by an additive manufacturing process.

10. A method of brazing a tube, the method comprising: positioning a probe in contact with a ring disposed at a joint between a first tube portion and a second tube portion, the probe biased toward the ring; positioning a heating element at the joint; melting the ring via operation of the heating element; and detecting movement of the probe along a probe central axis via a sensor operably connected to the probe, the movement of the probe indicative of melting of the ring; and stopping operation of the heating element when the sensor detects movement of the probe along the probe central axis.

11. The method of claim 10, wherein the probe is biased toward the ring via a compression spring operably connected to the probe.

12. The method of claim 10, wherein the heating element is one of a torch, a resistive heating element or an inductive heating element.

13. The method of claim 10, further comprising circulating a cooling fluid through one or more cooling channels of the probe to cool the probe.

14. The method of claim 13, wherein the cooling fluid is one of air or oil.

15. The method of claim 10, further comprising forming the probe from a metallic material.

16. The method of claim 10, wherein at least one of the first tube portion and the second tube portion are formed from aluminum.

17. The method of claim 10, wherein the first tube portion and the second tube portions are tubes of a heat exchanger.
